

End-of-outbreak decision-making under superspreading and onset-date delays

Methods and results for the 2018 Équateur Province (DRC) Ebola outbreak

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Scope

This document is the methods and results write-up for a study of how two modelling choices change the estimated risk of relaxing outbreak-control interventions:

- (i) the *mechanism* that generates heterogeneity in transmission — whether the excess variance attaches to a day, to a transmission event, or to an individual; and
- (ii) the routine practice of substituting a symptom-onset time series directly for an infection time series.

Five models are compared, all driven by the *same* onset-to-onset serial interval so that the comparison isolates mechanism and anchoring rather than confounding them with a different assumed interval. The output quantity throughout is the **risk of additional cases** (RAC): the posterior probability that at least one further case occurs after a given day, once the reproduction number is reset to its pre-intervention value. A companion quantity, the **risk of additional transmission** (RAT), is reported in a supplementary analysis, where the gap between the two measures the onset-versus-infection conflation directly.

There is deliberately no extended introduction and no discussion section at this stage. The sensitivity analyses that the design calls for are listed in Section 11 with their status; none of them has been run.

Every number below is generated from the files in `results/` by `scripts/run_report_numbers.py`; none is typed into the prose. The few measurements quoted from the validation studies are named with the file they come from.

Methods

1 Data and setting

The observed series is a daily count of Ebola virus disease cases by *symptom-onset date* for the 2018 Équateur Province outbreak in the Democratic Republic of the Congo: 54 cases, with onsets on 31 distinct days between 5 April 2018 and 2 June 2018, and at most 6 onsets on any one day. The series is padded with zero-count days out to the withdrawal of the Ebola Response Team (ERT), giving an analysis window of 111 days. Those trailing zeros are not padding in the cosmetic sense: they are the evidence that drives the risk down, and dropping them would leave the estimand undefined over exactly the period the decision is made in.

Day 0 is the first observed onset. Day indices and dates are

	Day index	Date
First onset (index case)	0	5 April 2018
ERT arrival	33	8 May 2018
Last observed onset	58	2 June 2018
ERT withdrawal (window end)	110	24 July 2018

with 28 cases before the ERT arrived and 26 after. No under-reporting adjustment is made.

The sole-index-case assumption. Day 0 is treated as an initial condition in every model, so the likelihood runs over days $1, \dots, 110$. Conditioning on a single imported case asserts that nothing was transmitting or incubating before 5 April 2018. For the infection-anchored models this is mild — the renewal recursion simply starts from one case. For the *onset-anchored* models it is a stronger assertion, because it also asserts an *empty incubation pipeline* at day 0: anyone infected shortly before the index case’s onset would contribute onsets in the first weeks that the model can explain only as transmission from the index case, inflating R_{pre} . The onset-anchored models are therefore the ones exposed to a mis-identified index case, and the initialisation sensitivity analysis of Section 11 is aimed at them.

Serial interval. Two estimates are available for this outbreak. The *generic* EVD estimate — discretised gamma with mean 15.30 d and standard deviation 9.30 d — is used throughout, for all five models. The *outbreak-specific* estimate, mean 19.46 d and standard deviation 6.08 d, fitted to contact-traced pairs from this outbreak, is retained only for a sensitivity analysis.

The reason for the choice is the variance budget of Section 2. Decomposing an onset-to-onset interval into an incubation period and a time-from-onset-to-transmission (TOST) delay requires the two variances to sum to the serial interval’s. The generic estimate has variance 86.49 d^2 , which comfortably admits standard published EVD incubation estimates; the outbreak-specific estimate has variance 36.97 d^2 , which is *smaller* than the incubation variance alone (65.61 d^2) and is therefore structurally inadmissible for the onset-anchored models. That incompatibility is itself informative about how much structure the decomposition imposes, and it is the reason the outbreak-specific sensitivity analysis can only be run for the naive models.

2 Delay distributions

All continuous delays are gammas, discretised by the method of Cori et al. (2013): the mass at integer lag s is $\int_{s-1}^{s+1} (1 - |s - u|) g(u) du$, normalised. Three discrete delays are used, and they differ in the lag they start at:

Array	Symbol	First lag	Role
Serial interval	w_s	1	onset of infector $\rightarrow$ onset of infectee
Incubation period	f_a^{inc}	1	infection $\rightarrow$ onset
Onset to transmission	f_s^{tost}	0	onset of infector $\rightarrow$ transmission

Write $F_r = \sum_{s=1}^r w_s$ with $F_0 = 0$. The lag-0 rule is what makes the onset-anchored recursion well ordered: f^{inc} must place no mass at lag 0, so that today’s onsets depend only on strictly earlier expected infections, while f^{tost} may, so that a case can transmit on its own onset day. At most one of the two may be supported at lag 0; otherwise infection, onset and onward transmission could all fall within a single day and the recursion would not be causal.

Decomposition and the variance budget. For an infector with onset on day u , transmission occurs on day $u + S$ with $S \sim f^{\text{tost}}$ and the infectee’s onset on day $u + S + A$ with $A \sim f^{\text{inc}}$. The onset-to-onset interval is therefore $S + A$, so matching the target serial interval requires

$$\mathbb{E}[S] + \mathbb{E}[A] = 15.30, \quad \text{Var}[S] + \text{Var}[A] = 86.49.$$

The incubation period is fixed from a published EVD estimate — mean 11.40 d, SD 8.10 d (WHO Ebola Response Team, 2014) — and f^{tost} is then the gamma carrying the residual mean and variance, here mean 3.90 d and SD 4.57 d. A negative residual variance would make the incubation estimate inadmissible; the implementation refuses it rather than proceeding.

Discretising and then convolving is not quite the same operation as convolving and then discretising, so the decomposition is checked numerically: the discrete convolution $f^{\text{tost}} * f^{\text{inc}}$ differs from the discretised target w by 0.00070 in total variation, against a configured tolerance of 0.02. All delays are truncated at lag 110, which is the analysis window itself, so nothing is truncated within it.

What the decomposition does and does not pin down. The convolution is faithful, but *how* the mean and variance are split between incubation and TOST is an assumption, and it is not identified by the serial interval alone. Since it is exactly this split that the onset-anchored models exploit, the sensitivity analysis over admissible incubation estimates in Section 11 bounds how much of the onset-anchoring result rests on the particular allocation used here.

3 Models

Let C_t be the observed count on day t . The three *naïve* models set $I_t := C_t$ — they read onset dates as infection dates, which is the practice under study. The two *onset-anchored* models set $D_t := C_t$ and treat onsets as onsets. R_t takes the value R_{pre} before the ERT arrives and R_{post} from its arrival onwards, and k is a dispersion parameter, held at a literature value in two analyses and estimated in the other two. $\text{NB}(\text{mean} = \mu, \text{disp} = \kappa)$ denotes the negative binomial with variance $\mu + \mu^2/\kappa$, and $\text{Gamma}(\alpha, \beta)$ has shape α and rate β .

3.1 Infection-anchored (naïve) models

DLO — day-level overdispersion.

$$I_t \sim \text{NB}\left(\text{mean} = R_t \sum_{s \geq 1} w_s I_{t-s}, \text{disp} = k\right). \quad (1)$$

The dispersion is *constant across days* and does not scale with the force of infection. That is the defining feature, and the reason the same numerical k means something quite different here than in SSE or SSI. No latent variables; the likelihood is closed-form.

SSE — superspreading events.

$$\lambda_t \sim \text{Gamma}(k\Lambda_t, k), \quad I_t \sim \text{Poisson}(R_t\lambda_t), \quad \Lambda_t = \sum_{s \geq 1} w_s I_{t-s}. \quad (2)$$

Marginalising λ_t gives $I_t \sim \text{NB}(\text{mean} = R_t\Lambda_t, \text{disp} = k\Lambda_t)$, so the dispersion *scales with the force of infection*: each infector independently contributes $\text{NB}(R_t w_\ell, k w_\ell)$ offspring at lag ℓ , and these pool by negative binomial closure. Because the marginal is available in closed form, SSE needs no latent variables — the only one of the five models with that property.

SSI — superspreading individuals.

$$Y_t | I_t \sim \text{Gamma}(kI_t, k), \quad I_t \sim \text{Poisson}\left(R_t \sum_{s \geq 1} w_s Y_{t-s}\right). \quad (3)$$

Each infected individual carries a latent infectivity with mean 1 and variance $1/k$; Y_t is the aggregate infectivity of the day- t cohort, and $I_t = 0$ forces $Y_t = 0$. Marginally the offspring distribution is $\text{NB}(R, k)$, the same as SSE — the two models differ only in mechanism. On this series SSI carries 31 latents.

3.2 Onset-anchored models

Transmission is anchored to the infector’s own onset through f^{tost} , and f^{inc} maps infections forward to onsets. This is appropriate for EVD, where pre-symptomatic transmission is negligible. In the inference-convenient forms below, E_t is the *expected* number of infections on day t ; the actual infection counts are marginalised out (Section 4).

All three share the same final step, the map from expected infections to observed onsets,

$$D_t \sim \text{Poisson}\left(\sum_{a \geq 1} f_a^{\text{inc}} E_{t-a}\right), \quad (4)$$

and differ only in how E_t is formed:

$$\text{Cori-SO :} \quad E_t = R_t \sum_{s \geq 0} f_s^{\text{tost}} D_{t-s}; \quad (5)$$

$$\text{SSE-SO :} \quad \tilde{\lambda}_t \sim \text{Gamma}\left(k \sum_{s \geq 0} f_s^{\text{tost}} D_{t-s}, k\right), \quad E_t = R_t \tilde{\lambda}_t; \quad (6)$$

$$\text{SSI-SO :} \quad Y_t | D_t \sim \text{Gamma}(kD_t, k), \quad E_t = R_t \sum_{s \geq 0} f_s^{\text{tost}} Y_{t-s}. \quad (7)$$

Cori-SO is the Poisson ($k \rightarrow \infty$) limit. It is *not* one of the compared models: it exists as the target of the collapse checks, alongside its infection-anchored counterpart.

The latent cost is asymmetric. SSI-SO carries one latent per day with an onset, the same 31 as SSI. SSE-SO carries one per day whose TOST-weighted onset sum is non-zero, and because the index case contributes $f_t^{\text{tost}} D_0 > 0$ at every lag out to the truncation, that is *every* inference day: 110 latents. Section 6 describes how they are handled.

3.3 The five models, and the switch convention

Model	Anchoring	Excess variance acts at the level of	Latents
DLO	infections (naive)	the day — NB on aggregate incidence	none
SSE	infections (naive)	the transmission event	none
SSI	infections (naive)	the individual	Y_t
SSE-SO	symptom onsets	the transmission event	$\tilde{\lambda}_t$
SSI-SO	symptom onsets	the individual	Y_t

When does R switch? R switches from R_{pre} to R_{post} on day 33 (8 May 2018) in **each model’s own time index**. In the onset-anchored models this is unambiguous: R_t multiplies the TOST-weighted force of infection producing *infections* on day t , so the intervention acts on transmission from that date and the resulting onsets appear an incubation period later. In the naive models R_t multiplies the force of infection producing *cases appearing* on day t , so the switch attributes the intervention’s effect to onsets from that date.

The displacement between the two conventions is the infectee’s **incubation period**, not the serial interval: a case with onset on day t was infected on day $t - A$ with $A \sim f^{\text{inc}}$, the naive model dates the intervention effect on that transmission to t and the onset-anchored model to $t - A$. The relevant delay is therefore $\mathbb{E}[A] = 11.40$ d. (The serial interval, 15.30 d, is the infector-onset to infectee-onset gap and does not separate the conventions.)

This asymmetry is deliberate and is a reportable consequence of onset-as-infection conflation rather than a modelling artefact: it is one of the mechanisms by which naive models mis-date an intervention’s effect, and Section 9 quantifies what it does to R_{post} and to the RAC curve. Section 11 lists the shifted-switch run that would separate this indexing effect from the structural one.

4 The two forms of the onset-anchored models are equivalent

The “natural” definitions of the onset-anchored models generate latent infection counts J_t and map them forward to onsets; the inference-convenient forms of Section 3 work directly with onsets. They define the same joint law for the observed onset process (D_t) .

In the natural form, with $\lambda_t = \sum_{s \geq 0} f_s^{\text{tost}} D_{t-s}$ (Cori-SO, SSE-SO) or $\lambda_t = \sum_{s \geq 0} f_s^{\text{tost}} Y_{t-s}$ (SSI-SO): (1) infections J_t are drawn from the model’s offspring distribution with mean $R_t \lambda_t$; (2) the J_u infections of day u are allocated to onset days by independent incubation draws, $(N_{u,a})_{a \geq 1} \mid J_u \sim \text{Multinomial}(J_u, (f_a^{\text{inc}}))$; (3) onsets are $D_t = \sum_{a \geq 1} N_{t-a,a}$.

4.1 The Poisson case (Cori-SO)

Let $E_t = R_t \lambda_t$, so $J_t \mid E_t \sim \text{Poisson}(E_t)$.

Step 1 — Poisson thinning. Multinomial splitting of a Poisson count yields *independent* Poisson parts: conditional on E_u , $N_{u,a} \sim \text{Poisson}(E_u f_a^{\text{inc}})$ independently over $a \geq 1$, with $J_u = \sum_a N_{u,a}$ recovered as their sum. The latent J_u therefore carries no information beyond the parts and can be dropped.

Step 2 — Poisson superposition. $D_t = \sum_{a \geq 1} N_{t-a,a}$ is a sum of independent Poissons, hence

$$D_t \mid (E_u)_{u < t} \sim \text{Poisson}\left(\sum_{a \geq 1} f_a^{\text{inc}} E_{t-a}\right), \quad (8)$$

which is exactly the inference-convenient form.

Step 3 — the joint law, not just the marginal. Because f^{inc} has no mass at lag 0, E_{t-a} for $a \geq 1$ is a function of $D_{0:t-1}$ only, so the display above is precisely the conditional law $p(D_t \mid D_{0:t-1})$. The parts $\{N_{u,a} : u + a = t\}$ composing D_t are disjoint from those composing any other $D_{t'}$, so distinct days’ onsets are conditionally independent given the E s. Chaining $p(D_{0:T}) = \prod_t p(D_t \mid D_{0:t-1})$ gives equality of the full joint laws. $\square$

4.2 SSE-SO

Step A — Poisson–Gamma mixture. If $\tilde{\lambda}_t \sim \text{Gamma}(k\lambda_t, k)$ and $J_t \mid \tilde{\lambda}_t \sim \text{Poisson}(R_t \tilde{\lambda}_t)$, then $R_t \tilde{\lambda}_t \sim \text{Gamma}(k\lambda_t, k/R_t)$ and, marginally,

$$J_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k\lambda_t), \quad (9)$$

which is the natural-form offspring law. The convenient form merely *un-marginalises* the negative binomial into an explicit per-day Gamma random effect.

Step B. Conditional on $\tilde{\lambda}_t$ (equivalently on $E_t = R_t \tilde{\lambda}_t$) the infection counts are Poisson, so Steps 1–3 above apply verbatim. The $\tilde{\lambda}_t$ are independent across t given (λ_t) , and each λ_t is a function of $D_{\leq t}$, so the day- t conditional still depends only on $D_{0:t-1}$ once $\tilde{\lambda}_{< t}$ is integrated out. $\square$

Consequence. The observed onsets are **not** marginally negative binomial: D_t pools contributions from many days, each carrying its own independent Gamma. There is no closed-form likelihood and the $\tilde{\lambda}_t$ must be sampled — in contrast to the infection-anchored SSE, which is latent-free.

4.3 SSI-SO

Identical to the Poisson case with $E_t = R_t \sum_{s \geq 0} f_s^{\text{tost}} Y_{t-s}$. The Gamma infectivities are already explicit in both forms, and the thinning and superposition arguments do not touch them. $\square$

4.4 What is tested

The equivalence is encoded as tests rather than left as an argument: setting $f_1^{\text{inc}} = 1$ (deterministic one-day incubation) and shifting indices recovers the infection-anchored models with $w_s = f_{s-1}^{\text{tost}}$; $k \rightarrow \infty$ collapses SSE-SO and SSI-SO onto Cori-SO; forward-simulating the natural form (explicit J_t plus a multinomial split) and the convenient form gives matching empirical distributions of $D_{1:T}$; and the convenient-form likelihood matches a brute-force marginalisation over small J_t on short synthetic histories.

5 The risk of additional cases and the risk of additional transmission

5.1 RAC is a retrospective reset posterior predictive

Writing the estimand as $P(\cdot \mid \text{data up to day } t)$ invites a filtering reading that the analysis does not use. It is stated instead as a procedure:

Fit $(R_{\text{pre}}, R_{\text{post}}, k)$ and all latent states to the complete outbreak record, days 0–110. Then, for each day t : retain the inferred state attached to the observed history through day t , discard the realised trajectory after day t , reset R to R_{pre} from day $t+1$ onwards, and simulate a counterfactual future. **RAC(t) is the posterior probability that this replicated future contains at least one further case.**

Two things follow that the shorthand hides. The fit is fully retrospective by design: the parameters come from all 111 days, not from the history through t . And for the latent-variable models the *retained state* also comes from the full-data posterior, so it is a *smoothed* state informed by data after day t — see Section 5.5.

RAC(t) is one number per day, not a distribution: it is a posterior probability, so the average over draws happens inside it, and what is reported is $1 - \exp(\log P(\text{no further case}))$ over the posterior draws. Because it is an average it carries Monte-Carlo error, which is computed from the between-chain spread and written beside every curve; no curve is published without it.

5.2 Closed forms for the naive models

Define the pooled remaining-transmission weight

$$\Lambda(t) = \sum_{u \leq t} I_u (1 - F_{t-u}) \quad (\text{SSE, DLO}), \quad \Lambda_Y(t) = \sum_{u \leq t} Y_u (1 - F_{t-u}) \quad (\text{SSI}), \quad (10)$$

where $(1 - F_{t-u})$ is the probability that a case first appearing on day u still has transmission to come after day t ; the $u = t$ term enters with full weight. Then

Model	$P(\text{no further cases after day } t)$
SSE	$(1 + R_{\text{pre}}/k)^{-k\Lambda(t)}$
SSI	$\mathbb{E}_{\text{posterior}}[\exp(-R_{\text{pre}}\Lambda_Y(t))]$
DLO	$\prod_{j>t} (1 + R_{\text{pre}}\mu_j/k)^{-k}, \quad \mu_j = \sum_{s \geq 1} w_s I_{j-s} \text{ with } I_u := 0 \text{ for } u > t$

The DLO product is finite in practice, since $\mu_j = 0$ once j exceeds t plus the support of w .

5.3 Why DLO is structurally different

SSE and SSI depend on the observed history *only through the scalar* $\Lambda(t)$: the total remaining offspring pools into a single $\text{NB}(R_{\text{pre}}\Lambda, k\Lambda)$ or, given the latents, $\text{Poisson}(R_{\text{pre}}\Lambda_Y)$. DLO does

not. Its risk depends on the whole temporal *profile* $(\mu_j)_{j>t}$ of the future force of infection, because a fresh dispersion k that does not scale with the force of infection is applied on each future day. Writing $\varphi(\mu) = -k \log(1 + R\mu/k)$, which is convex with $\varphi(0) = 0$ and hence superadditive,

$$-R\Lambda(t) \leq \log P(\text{no further cases})_{DLO} = \sum_{j>t} \varphi(\mu_j) \leq \varphi(\Lambda(t)). \quad (11)$$

Spreading the same total $\Lambda(t)$ across days *dilutes* the overdispersion, so a value of k calibrated as an individual-level offspring dispersion does far less work in DLO than in SSE or SSI.

The bounds in (11) hold always; the *direction* of the comparison with SSE does not, and is a property of this outbreak's thin, spread-out future force of infection rather than a theorem. Concentrating $\Lambda(t)$ into a day or two puts DLO below SSE instead. The implementation and its tests are written against the bounds.

5.4 The onset-anchored reset state, and RAT

Under the onset-anchored models the observed onsets do not determine the state needed to project forward. At day t there is a latent **incubation pipeline** of individuals already infected but not yet symptomatic, of size

$$\text{Poisson} \left(\sum_{u \leq t} E_u \left(1 - \sum_{a \leq t-u} f_a^{\text{inc}} \right) \right) \quad (12)$$

given the E s, allocated to future onset days by the residual incubation distribution $f_a^{\text{inc}}/(1 - \sum_{a' \leq t-u} f_{a'}^{\text{inc}})$ for $a > t-u$. The implementation constructs this explicitly. Restarting the recursion from the observed onsets alone would silently drop the pipeline and understate the risk; this is an implementation requirement of the estimand, not an ambiguity in it.

With $H(t)$ the contribution of future transmission from retained onset cohorts, $M(t)$ the mean size of the retained pipeline and c the expected onward transmission a pipeline individual generates, the conditional zero-probabilities are evaluated analytically as

$$\log P(\text{no further case}) = -H(t) - M(t), \quad (13)$$

$$\log P(\text{no further transmission}) = -H(t) - M(t)(1 - e^{-c}), \quad (14)$$

and averaged over the posterior draws. Hence:

- **RAC**(t) — the probability of at least one further *case* in the onset series after day t . It includes pipeline individuals who become symptomatic after t even if no new transmission occurs, so it does not fall towards zero until the pipeline has drained. This is the directly observable quantity and the headline curve everywhere.
- **RAT**(t) — the probability of at least one further *transmission event* after day t . It isolates the effect of relaxing interventions but is not directly observable.

$\text{RAC}(t) \geq \text{RAT}(t)$ always, and the gap is exactly the pipeline contribution. Under the three naive models the two quantities coincide *by assumption* — which is precisely the conflation this study is about — so RAT is reported only in the supplementary analysis of Section 10.

The SSE-SO latent at the final day of the window needs the same treatment even though it can reach no fitted onset: incubation starts at lag 1, so its infections belong entirely to the reset pipeline, and its conditional law is its Gamma prior.

5.5 Latents at the conditioning day: the smoothing approximation

For the latent-variable models, RAC at day t in principle wants latents conditioned on the history through t — a *filtering* distribution — whereas a single full-data fit delivers the *smoothed* distribution given the whole record.

The full-data (smoothed) posterior is used for every t . One fit per model then serves every day of every curve, which keeps each analysis to a handful of MCMC runs rather than several hundred. This is consistent with the analysis being fully retrospective: the parameters are estimated from all the data by design, and the latents are treated the same way. The consequence must be stated plainly — the projection at day t uses latents informed by data after day t , so $\text{RAC}(t)$ is not literally a probability conditioned on the history through t for SSI, SSE-SO and SSI-SO. DLO and SSE are unaffected: they have no latents.

The *direction* of the effect is a hypothesis, not an assertion. Smoothing sharpens the latents but does not push them uniformly one way: post- t zeros are evidence that recent cohorts had low infectivity and pull the retained state down, while post- t positive observations are evidence that earlier cohorts had high infectivity and pull it up. After the final onset every subsequent observation is a zero, so the expectation is a predominantly downward effect over the decision-relevant late period, and an ambiguous one earlier.

That hypothesis has been measured rather than assumed, using the particle filter’s forward pass, which produces filtering distributions as a by-product. Both signs appear, and where predicted. For SSI at a fixed parameter vector, smoothed minus filtering RAC is $+0.64$ around day 2 — where the filter has not yet seen the cases that reveal a high early infectivity — and negative after the last observed case, mean -0.031 and worst -0.153 around day 70 (`validation/results/rac_validation.md`). For the onset-anchored models the same comparison, with the opposite sign convention, gives filtering minus smoothed RAC of about -0.63 to -0.66 at its largest early and a mean of about $+0.013$ after the final onset (`validation/results/onset_particle_mcmc_validation.md`). The late-period effect is small enough not to move the threshold crossings by more than about a day, but it is a signed, measured number rather than a claim.

5.6 Reconstructing the latents the fit integrated out

The latent parameterisation of Section 6 integrates out, in closed form, every latent that the data constrain only through the factor $\exp(-\sum_j \mu_j)$. Those latents are not in the posterior draws, and on this series they are exactly the days from the last observed onset onwards — precisely the days a late conditioning day t needs for its incubation pipeline. Reading the retained state straight off the posterior array would look plausible and be truncated at day 58, understating RAC for exactly the late days where the decision is made.

The removed latents are therefore rebuilt from their exact conditional $\text{Gamma}(k \cdot \text{scale}_u, k + c_u)$ before any risk is computed, and spliced onto the sampled block to give a complete latent path by day. This is exact — the removed latents are conditionally independent of the sampled ones as well as of each other — and it costs one Gamma draw per removed latent per posterior draw, done once rather than once per t . The reconstruction is validated end to end by a matched pair of fits: the default parameterisation, which needs the reconstruction, against one that samples every latent, which does not. Their RAC curves are indistinguishable, differing by at most 2.4×10^{-3} — the same spread as two runs of the *same* fit (`validation/results/rac_validation.md`).

6 Inference

6.1 Sampling

All fits are run in PyMC with NUTS. Each analysis uses 4 chains of 2000 draws after 2000 tuning iterations, at a target acceptance probability of 0.90 for the first three analyses and 0.95 for the estimated- k onset-anchored analysis. Seeds live in the configuration file. One fit per (model, analysis) pair jointly estimates R_{pre} , R_{post} , any latents, and — in Analyses 2 and 4 — k .

SSE and SSI appear in both the naive and the onset-anchored analyses. They are refitted independently under each analysis’s own seed, so their summaries can differ between figures by Monte-Carlo error; no number is carried across.

6.2 Priors

Priors are proper and identical across the models within an analysis. This is not a stylistic choice: the posterior model probabilities are Bayes factors, and diffuse or improper priors would make them arbitrary.

- R_{pre} and R_{post} : independent log-normals with median 1.0 and 95% interval (0.2, 5.0).
- k , where estimated: a deliberately fairly informative log-normal with median 0.18 and 95% interval (0.09, 0.36), the same in every model.
- Prior model probabilities: uniform over the models in each analysis, i.e. 1/3 in Analyses 1–2 and 1/4 in Analyses 3–4.

Identical priors are necessary but not sufficient for a fair comparison. Propriety makes the Bayes factors well defined; it does not make them *neutral* when the same symbol means different things in different models. k is exactly such a symbol: an individual-level offspring dispersion in SSE and SSI, a day-level incidence dispersion in DLO (Section 5.3). Placing the same prior on both is a deliberate choice, and the model-probability panel of Fig. 1 in particular compares models one of which is *deliberately miscalibrated*, in order to demonstrate the error under study. It is evidence about those specified models, not clean evidence about mechanism. Prior-predictive calibration is out of scope here, so this caveat carries the weight instead.

6.3 The latent block

Sampling the latent Gammas directly is where NUTS struggles, badly: the SSE-SO shapes on this series span several orders of magnitude down to $\sim 10^{-6}$, and a Gamma coordinate that small has a typical value outside the range of double precision, so it cannot be tuned into behaving. Two mechanisms are used, and they compose.

- **Exact marginalisation.** A latent that reaches no observation day carrying a case enters the likelihood only through $\exp(-\sum_j \mu_j)$, which is linear in the latents and so factorises. Its Gamma prior is conjugate to that, giving $\int \text{Gamma}(y; k \text{ scale}, k) e^{-cy} dy = (1 + c/k)^{-k \text{ scale}}$ and a conditional posterior $\text{Gamma}(k \text{ scale}, k + c)$. On this series that removes 52 of SSE-SO’s 110 latents, leaving 58 sampled — exactly the pathological tail, since the last case is on day 58 — and 1 of SSI’s and SSI-SO’s 31. It is an exact operation, and Section 5.6 describes how the removed latents are recovered where the risk calculators need them.
- **Inverse-CDF reparameterisation.** The surviving latents are sampled as $\text{Uniform}(0, 1)$ variates pushed through the Gamma quantile function, which gives a bounded, flat-prior, constant-geometry parameter space and does have a gradient in the PyTensor in use, so it runs under NUTS.

The choice rests on a benchmark of 74 MCMC runs (`validation/results/sampler_benchmark.md`). Its main negative finding is worth stating because it constrains what else could be tried: this is a *geometry* problem, not a warm-up problem. Mean-1 rescaling of the latents leaves the real SSE-SO model with 3999 of 4000 draws divergent, exactly as the untouched model does, and more tuning makes $\hat{R}$ worse rather than better — rescaling is a translation on the log scale, so it leaves the curvature untouched. The inverse-CDF reparameterisation takes the same model from 4000 divergences to none.

Convergence. Across all models of all four analyses, the worst $\hat{R}$ for the reported scalars is 1.00, the smallest bulk effective sample size is 3575, and the total number of divergent transitions is 0. Per analysis, the minimum bulk effective sample sizes are 4855 (Analysis 1), 4074 (Analysis 2), 4190 (Analysis 3) and 3575 (Analysis 4), with no divergences in any of them.

6.4 Model evidence and posterior model probabilities

The marginal likelihood $p(D_{1:110} \mid \text{model})$ is computed with the parameters *and* the latents integrated out, from the built model’s own joint log-density rather than from a re-derivation of the likelihood, so that the integral estimated is the one that was sampled. The integral is done in PyMC’s unconstrained coordinates with the Jacobian included; getting the transform’s direction wrong would shift every evidence by a constant — a perfectly plausible number and a wrong Bayes factor — so that is pinned by test rather than trusted. Bridge sampling is the estimator; prior Monte Carlo and importance sampling are kept as agreement checks, and deterministic quadrature as the exact answer where the space is small enough.

Three checks establish that the numbers are right rather than merely self-consistent (`validation/results/evidence_validation.md`): bridge sampling reproduces deterministic quadrature for DLO and SSE on the real series to 1.1×10^{-3} and 2.9×10^{-3} nats, with the quadrature box’s boundary density 41–44 nats below its peak; the three estimators agree on SSI, all within their own error bars; and the evidence is invariant to the latent parameterisation, which is what shows the marginalisation carries the whole removed factor rather than just its shape.

6.5 Comparing the dispersion posteriors

Where k is estimated, the models are handed the *same* prior on it, so wherever their posteriors end up apart is a measurement rather than a demonstration. Three complementary measures are reported for every pair, because none says enough alone: the **median ratio** (location, but not width); the **overlap** $\int \min(p_a, p_b)$ (scale-free, symmetric, and finite where a Kullback–Leibler divergence would not be); and $P(k_a > k_b)$ for independent draws. All are computed on the log scale, as the density panels are drawn; the overlap is invariant to that choice. Every pair is compared rather than a designated reference model.

6.6 Independent validation of the pipeline

Two particle-filter checks answer different questions and are kept separate. At fixed $(R_{\text{pre}}, R_{\text{post}}, k)$, with conditioning matched to the pipeline, the filter is an equality check on the RAC arithmetic; agreement is required and is obtained, to 3.2×10^{-3} for SSI and 0.021 for the onset-anchored models. Over the parameters, particle marginal Metropolis–Hastings gives an independent route to the posterior that shares no machinery with the PyMC fits — the main guard against the latent-parameterisation risk, which simulation-based calibration could not catch, since it validates an implementation against itself. PMMH and PyMC 95% intervals overlap for all six synthetic parameter comparisons, and the largest measured variance of the estimated log-likelihood is 0.081 at 250 particles, comfortably inside the range PMMH tolerates. A PMMH RAC curve should *not* match the MCMC one for the latent models: the filter conditions on filtering latents where the pipeline uses smoothed ones, and that gap is the Section 5.5 measurement rather than a defect.

Externally, the Poisson-limit RAC reproduces the published Équateur risk curve of Thompson et al. (2024) exactly under their conventions, which differ from this project’s by exactly one day: their published R posterior for this outbreak is recovered, and the two risk curves agree to 2.2×10^{-4} over 200 000 draws (`validation/results/rac_validation.md`). Particle MCMC remains a check throughout and never a results path; the main analyses stay in PyMC.

Results

Four analyses are reported. Analyses 1 and 2 compare the three naive models, with k held at the literature value 0.18 and then estimated; Analyses 3 and 4 add the onset-anchored models to SSE and SSI, again fixed and then estimated. Throughout, “the crossing” means the first day on which a curve falls below a threshold *and stays there* — the same definition the markers on the

figures use, and computed by the same code. Reference values quoted at day 90 (4 July 2018) are read from the fitted curves partway down their descent, where the curves are furthest apart.

7 Analysis 1 — naive models at a fixed literature k

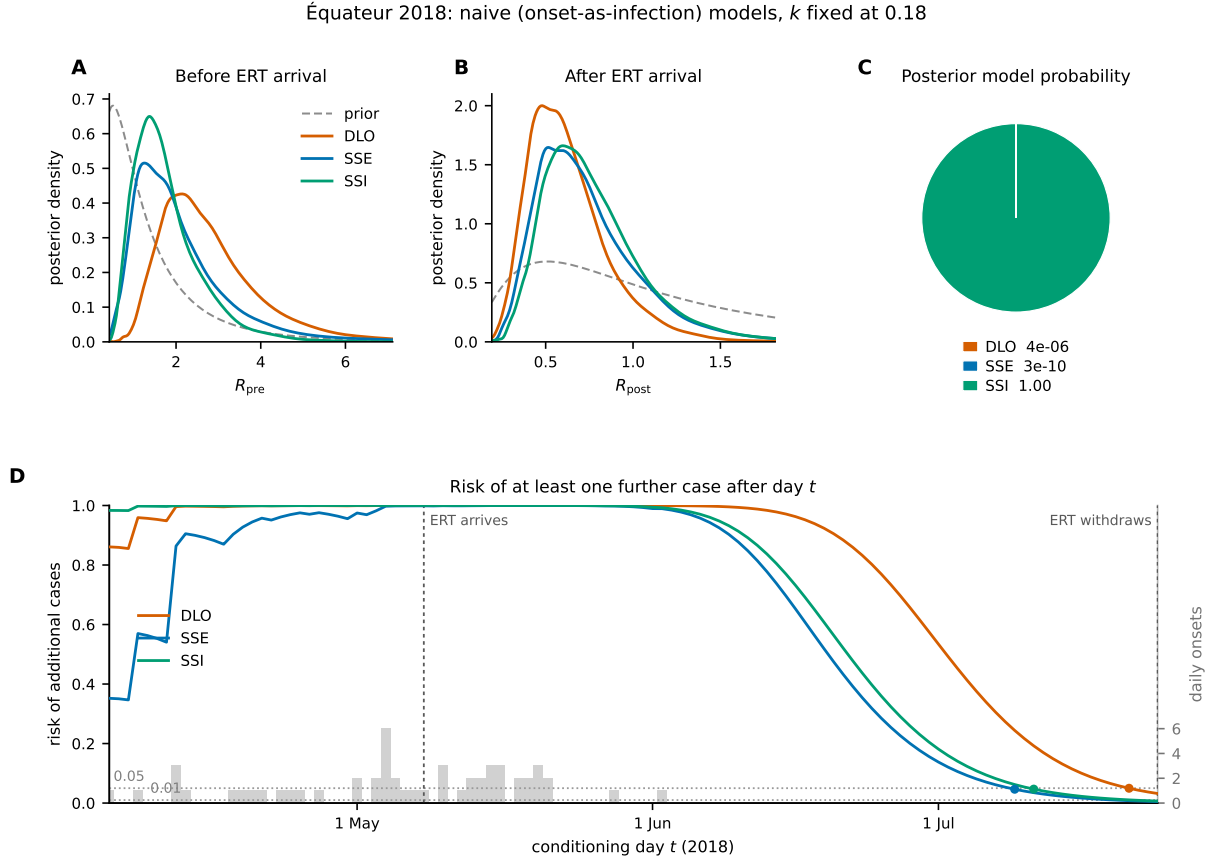


Figure 1: The mechanism of overdispersion changes the end-of-outbreak risk, at a dispersion parameter that is only meaningful for two of the three models. DLO, SSE and SSI fitted to the Équateur onset series with k fixed at 0.18. **A**, **B**: posterior densities for R_{pre} and R_{post} , over the shared prior. **C**: posterior model probabilities under a uniform prior over the 3 models; note that one of the three is deliberately miscalibrated (Section 6), so this panel is evidence about the models *as specified*, priors included. **D**: RAC against conditioning day, with daily onsets behind it, the ERT arrival and withdrawal marked, and the 0.05 and 0.01 decision thresholds drawn. The marker on each curve is its 0.05 crossing.

- **The same k buys DLO almost no overdispersion.** At day 90 the RAC is 0.413 under DLO against 0.093 under SSE and 0.123 under SSI — a factor of more than four between models that were handed the identical dispersion parameter. This is the dilution argument of Section 5.3 measured: the future force of infection after the last case is thin and spread over the whole serial interval, which is precisely the regime in which a day-level k does almost nothing.
- **The consequence for a declaration is weeks, not days.** DLO's RAC first settles below 0.05 on day 107 (21 July 2018) against day 95 for SSE and day 97 for SSI. At the 0.01 threshold DLO **never** settles within the analysis window, while SSE does so on day 106 and SSI on day 108. On the day the ERT actually withdrew, DLO still puts 0.032 on a further case against 0.005 and 0.007.
- **SSI sits on the same side of SSE as DLO does, and that is not a bug.** After the final case SSI's RAC is *above* SSE's (0.123 against 0.093 at day 90), which reads backwards

until one asks what each model’s heterogeneity is attached to. SSI’s is a property of *people who have been watched*: 54 known individuals have now gone weeks without infecting anyone, which is direct evidence they were not the superspreading kind, and their inferred residual infectiousness is revised down accordingly. SSE’s is a property of *events that have not happened yet*, about which a run of zeros says nothing, so it gets no such revision. At matched expected further cases, SSE’s residual risk is a rare burst and SSI’s a more likely singleton — and RAC asks only “any further case?”. **The report should not be read as “DLO is the odd one out”**: DLO and SSI are on the same side of SSE, for related reasons. The axis is how many units the same $\Lambda(t)$ is divided over, and whether each unit’s dispersion survives to time t : DLO divides over future days, SSI over past individuals, SSE not at all.

- **The ordering is a property of the question as much as of the models.** Asking instead “would transmission re-establish?” reverses SSE and SSI on this same data, because a cluster seeds a new chain far more readily than a lone case. Both orderings are correct answers to different questions, and the end-of-outbreak criterion in use here asks the first one.
- **The model probabilities strongly favour SSI** — 1.00 against 4×10^{-6} for DLO and 3×10^{-10} for SSE, from log evidences of -84.8 , -97.1 and -106.8 nats. Read this with the caveat above: at a fixed k the panel is partly reporting that 0.18 is a better value for SSI than for the other two, which is aim (i) restated rather than independent evidence about mechanism. Analysis 2 separates the two readings.
- **R_{post} is comparable across the three models** — 0.59 (0.30–1.20), 0.68 (0.34–1.48) and 0.72 (0.36–1.48) for DLO, SSE and SSI (median and 95% credible interval) — so the divergence in the risk curves is not coming from a different inferred effect of the intervention. R_{pre} is 2.55 (1.20–6.05), 1.81 (0.72–5.35) and 1.61 (0.73–3.87).
- **Monte-Carlo error is not what separates the curves.** Over the case-free tail where the crossings happen, the largest Monte-Carlo standard error on any of the three curves is 0.0015, two orders of magnitude below the differences above.

8 Analysis 2 — naive models with k estimated

- **The models disagree about k , but not along the axis one might expect.** The posterior medians (95% CrI) are 0.378 (0.230–0.629) for DLO, 0.502 (0.307–0.853) for SSE and 0.141 (0.072–0.271) for SSI. DLO against SSI: overlap 0.094, median ratio 2.69, $P(k_{\text{DLO}} > k_{\text{SSI}}) = 0.990$. SSE against SSI: overlap 0.032, ratio 3.57, probability 0.999. But **DLO against SSE: overlap 0.589** and $P(k_{\text{DLO}} > k_{\text{SSE}}) = 0.219$ — the two are barely distinguishable. **SSI is the odd one out, not DLO.**
- **The split is day-level against individual-level, which is where onset-as-infection bites hardest.** DLO and SSE both attach their excess variance to a *day*: DLO to the day’s aggregate incidence, SSE to the day’s pooled transmission through a freshly drawn λ_t . SSI attaches it to the *individual*. The series being fitted is onsets, and an incubation period of mean 11.40 d and SD 8.10 d scatters each day’s infections forward over a wide kernel — so a model reading onsets as infections sees day-to-day variation that the convolution has already largely averaged out, and answers with a larger k (less overdispersion) than the infection process carried. SSI is close to invariant to this, because the convolution *regroups* individuals across days and the aggregate infectivity of n i.i.d. individuals is $\text{Gamma}(kn, k)$ whichever n they are: regrouping moves the cohorts, not the variance. This is a mechanism proposed for the measurement, not a second measurement — Analysis 2 has no non-naive arm, so it cannot separate this from the purely structural difference of Section 5.3. Analysis 4 is where it is tested.
- **Only SSI’s posterior is compatible with the literature value.** SSI’s 95% credible

Équateur 2018: naïve (onset-as-infection) models, k estimated

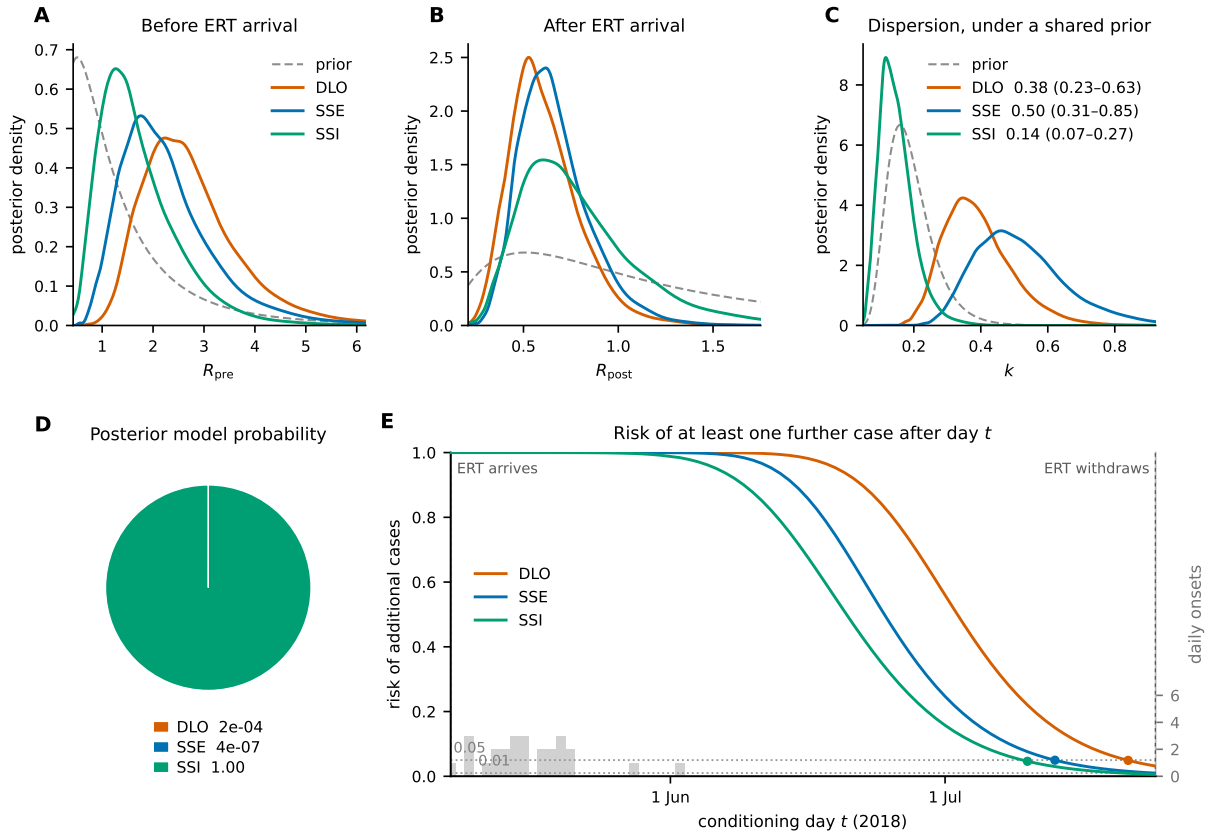


Figure 2: Estimating k under a shared prior shows that the models want different values of it — and that the risk gap survives anyway. As Fig. 1, with k estimated. **C:** the three k posteriors over the prior they share, with each model's posterior median and 95% credible interval in the legend. **D:** posterior model probabilities. **E:** RAC, with the view starting at the ERT's arrival because all three curves sit flat at 1 until well past the last case.

interval, (0.072, 0.271), contains 0.18; DLO's (0.230, 0.629) and SSE's (0.307, 0.853) do not, and both medians sit above the prior's own 97.5th percentile of 0.36, despite that prior being deliberately informative.

- **The evidence says the same thing more sharply.** Letting k move is worth +7.41 nats to SSE and +3.90 to DLO, but only +0.26 to SSI. Analysis 1 was charging DLO and SSE for a value that was never theirs, and 0.18 really is SSI's number on this series.
- **The risk gap survives, which is the point of the Fig. 1 / Fig. 2 pair.** Each model now sits at its own best k , and DLO still crosses 0.05 on day 107 against day 99 for SSE and day 96 for SSI, and still **never** reaches 0.01 inside the window, where SSE does on day 110 and SSI on day 107. Misapplying a literature k makes the gap; the day-level mechanism keeps it.
- **The posterior model probabilities are unchanged in direction** — SSI 1.00, DLO 2×10^{-4} , SSE 4×10^{-7} — and now carry a cleaner reading, since each model has been allowed its own dispersion.

9 Analyses 3 and 4 — onset anchoring

Équateur 2018: onset anchoring compared, k fixed at 0.18

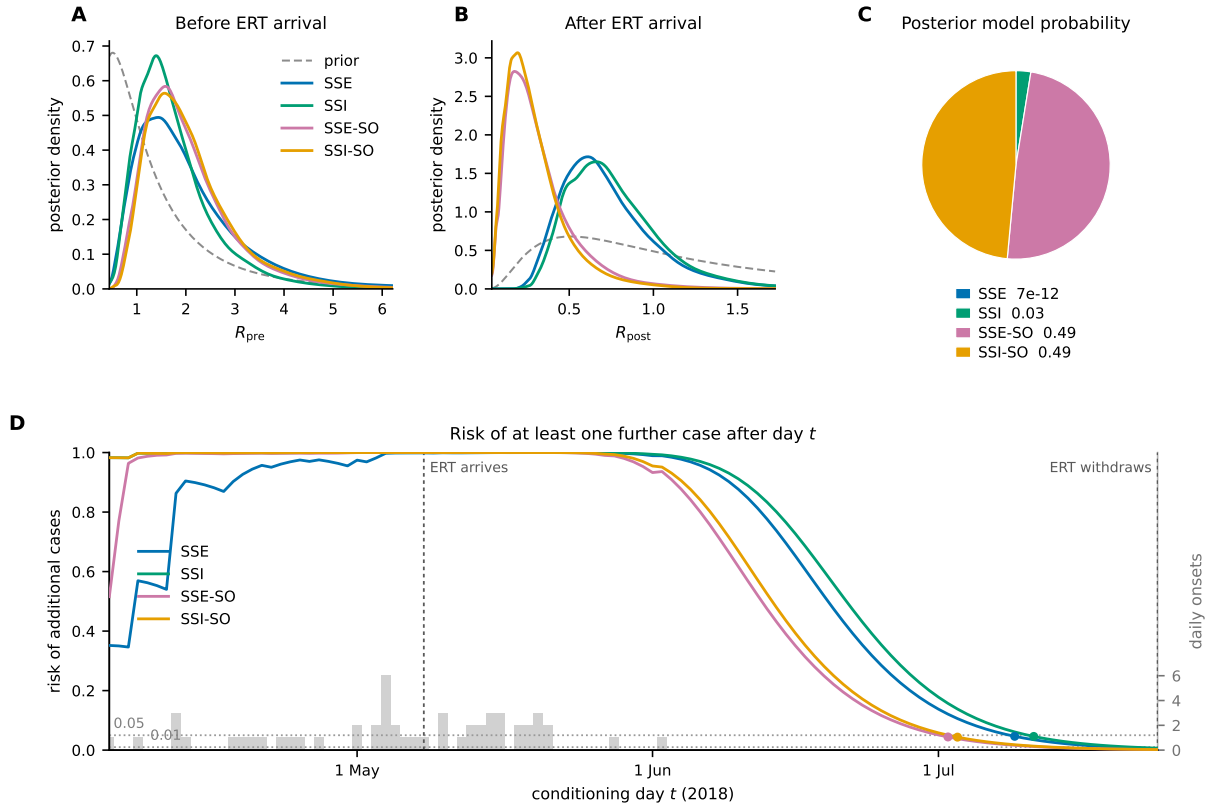


Figure 3: Onset anchoring at a fixed k . SSE, SSI, SSE-SO and SSI-SO with k held at 0.18; panels as in Fig. 1, with a uniform prior over 4 models in C. All four models are driven by the same onset-to-onset serial interval, so the comparison isolates anchoring.

- **Onset anchoring advances both RAC crossings by 8–12 days, and this is the headline.** With k estimated (Fig. 4), SSE-SO crosses 0.05 on day 87 against SSE's day 99 — an advance of 12 days — and 0.01 on day 98 against day 110, an advance of 12 days. SSI-SO advances SSI's crossings by 8 and 8 days respectively (day 88 against day 96, and day 99

Équateur 2018: onset anchoring compared, k estimated

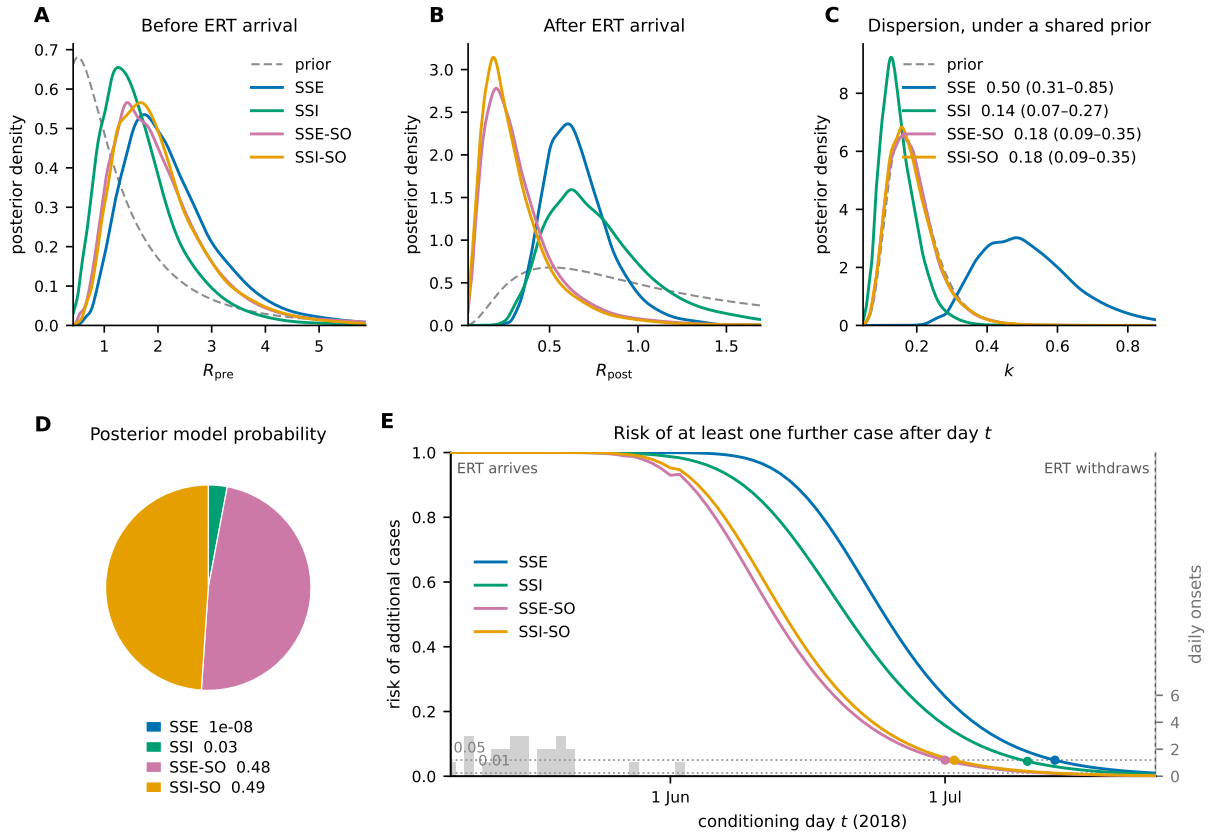


Figure 4: Onset anchoring with k estimated — the test of the Analysis 2 prediction. Panels as in Fig. 2. **C** is where the attenuation hypothesis is settled: SSE-SO's k posterior should sit below naive SSE's if the incubation convolution is what inflates day-level dispersion estimates, while SSI-SO's should barely move from SSI's.

against day 107). At a fixed k (Fig. 3) the same comparisons give 7 and 7 days for SSE-SO against SSE, and 8 and 9 days for SSI-SO against SSI. The displacement is on the scale of the mean incubation period, 11.40 d.

- **Why the direction is *earlier*, and why it is not a shorter generation time.** All four models share the same onset-to-onset serial interval; what changes is which side of the conditioning day a transmission falls on once that interval is cut into transmission and incubation pieces. The onset-anchored models place infection at transmission time, so infections already generated under the historical R_{post} stay in the incubation pipeline when controls are reset, and the remaining TOST opportunity of older onset cohorts is already largely spent. The naive models move that infection decision to the later symptom-onset date and apply the reset R_{pre} there, which makes the residual state look younger and drain later.
- **The same indexing error inflates naive R_{post} threefold.** With k estimated, the posterior medians (95% CrI) are 0.63 (0.37–1.11) for SSE and 0.73 (0.35–1.60) for SSI, against 0.28 (0.08–0.95) for SSE-SO and 0.26 (0.08–0.89) for SSI-SO. The naive convention applies the day-33 switch to *onsets*, so post-arrival onsets that arose from pre-arrival infections are charged to the controlled period. R_{pre} is much more stable across the four — 2.09 (0.98–4.77), 1.54 (0.65–3.67), 1.86 (0.84–4.61) and 1.87 (0.87–4.41) — which is what one expects if the effect is about *when* the switch lands rather than about the epidemic’s early growth.
- **The Analysis 2 attenuation prediction passes.** SSE-SO’s k posterior, 0.178 (0.088–0.351), sits below naive SSE’s 0.501 (0.306–0.847) and towards the literature value: median ratio 2.81, overlap 0.089, $P(k_{\text{SSE}} > k_{\text{SSE-SO}}) = 0.992$. SSI moves much less — 0.140 (0.072–0.271) against SSI-SO’s 0.175 (0.088–0.353), overlap 0.750. Incubation convolution attenuates *day-level* dispersion when onsets are treated as infections; individual-level dispersion is comparatively stable to regrouping, exactly as predicted.
- **Once anchoring is made explicit, the mechanism stops mattering for k .** SSE-SO and SSI-SO agree almost exactly — overlap 0.980, median ratio 1.02, $P(k_{\text{SSE-SO}} > k_{\text{SSI-SO}}) = 0.505$ — where their naive counterparts differ by a factor of 3.58 at an overlap of 0.034. The event-versus-individual gap in Analysis 2 was to a large extent the conflation, not the mechanism.
- **The onset-anchored models are strongly preferred, and the two of them are indistinguishable.** With k estimated, the posterior model probabilities are 0.48 for SSE-SO and 0.49 for SSI-SO, against 0.03 for SSI and 1×10^{-8} for SSE; the log evidences are -81.8 , -81.8 , -84.6 and -99.4 nats. The fixed- k analysis gives the same ordering (0.49, 0.49, 0.03, 7×10^{-12}).
- **The onset-anchored models gain nothing from estimating k , which is a consistency check on the previous two bullets.** Releasing k is worth $+0.01$ nats to SSE-SO and $+0.04$ to SSI-SO — within their own estimation error — against $+7.41$ for naive SSE. Their k posteriors already sit on 0.18, so fixing it there costs them nothing.
- **Do not universalise the direction of the 8–12-day advance.** It is measured under this reset convention, this switch convention, this outbreak history and this posterior. The incubation-scale explanation is the mechanism for *this* result, not a theorem for arbitrary time series or interventions.

10 Supplementary — RAC against RAT, and the incubation pipeline

- **The pipeline is worth about a quarter of a unit of probability at its widest.** With k estimated, RAC exceeds RAT by at most 0.269 for SSE-SO, on day 64 (8 June 2018), and by at most 0.240 for SSI-SO, on day 65. The gap peaks shortly after the last observed

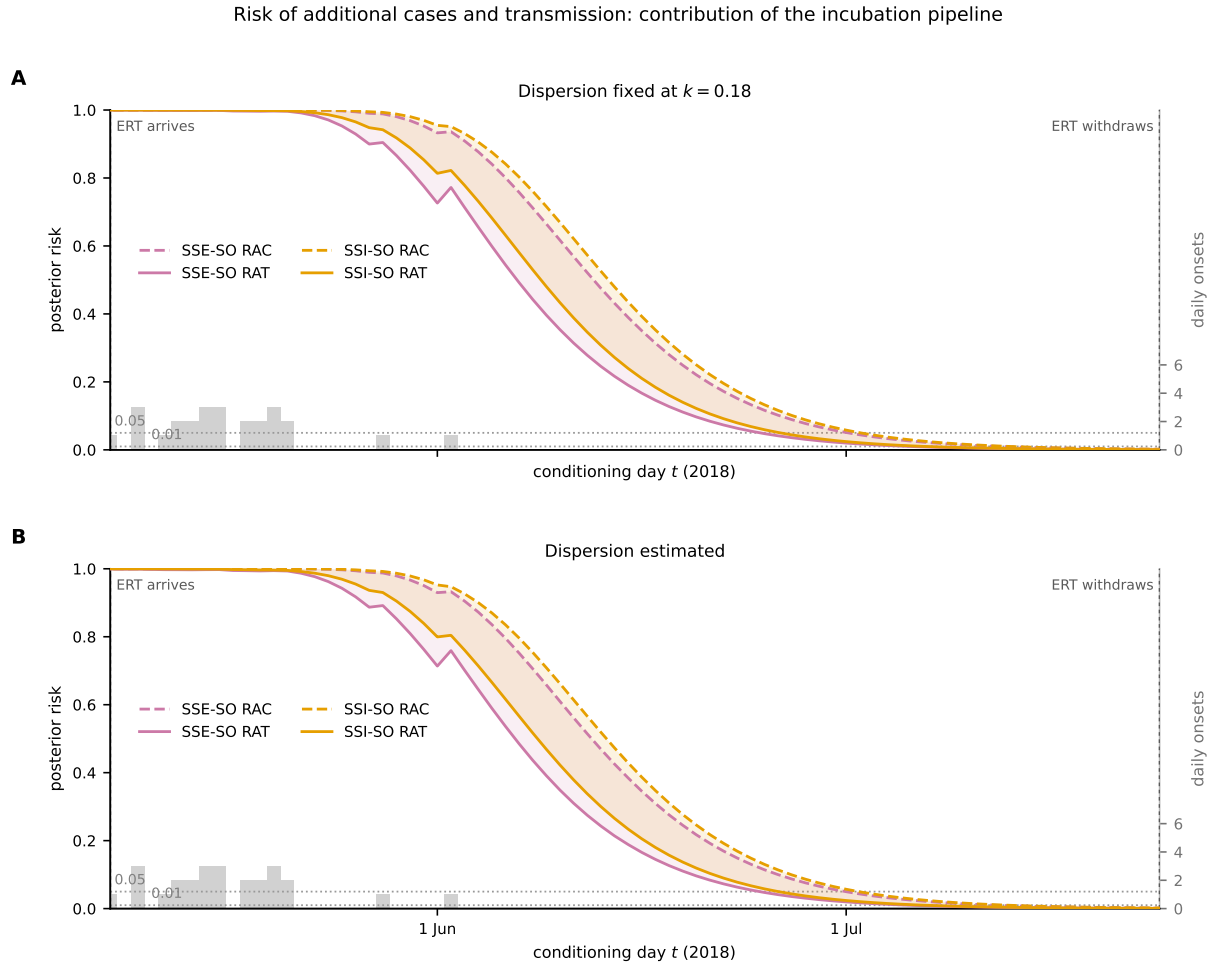


Figure 5: The gap between the risk of additional cases and the risk of additional transmission is the incubation pipeline, quantified. SSE-SO and SSI-SO with k fixed at 0.18 (A) and estimated (B). Dashed: RAC. Solid: RAT. The shaded region between them is the contribution of individuals already infected on the conditioning day but not yet symptomatic. Under the three naive models these two curves coincide by assumption, which is the conflation being measured.

Table 1: RAC threshold crossings across the four analyses: the first day on which the curve settles below the threshold, with the calendar date. The ERT withdrew on day 110 (24 July 2018); “never” means the curve had not settled below the threshold by then.

Analysis	Model	RAC < 0.05		RAC < 0.01	
		day	date	day	date
1: naive, $k = 0.18$	DLO	107	21 July 2018	never	never
	SSE	95	9 July 2018	106	20 July 2018
	SSI	97	11 July 2018	108	22 July 2018
2: naive, k estimated	DLO	107	21 July 2018	never	never
	SSE	99	13 July 2018	110	24 July 2018
	SSI	96	10 July 2018	107	21 July 2018
3: onset, $k = 0.18$	SSE	95	9 July 2018	106	20 July 2018
	SSI	97	11 July 2018	108	22 July 2018
	SSE-SO	88	2 July 2018	99	13 July 2018
	SSI-SO	89	3 July 2018	99	13 July 2018
4: onset, k estimated	SSE	99	13 July 2018	110	24 July 2018
	SSI	96	10 July 2018	107	21 July 2018
	SSE-SO	87	1 July 2018	98	12 July 2018
	SSI-SO	88	2 July 2018	99	13 July 2018

onset (day 58), which is when the pipeline is at its largest relative to the transmission still to come, and closes as it drains.

- **It delays a case-based declaration by about a working week.** With k estimated, SSE-SO’s RAT settles below 0.05 on day 81 and below 0.01 on day 92, against day 87 and day 98 for RAC — later by 6 and 6 days. For SSI-SO the same comparison gives 5 and 6 days (RAT on days 83 and 93 against RAC on days 88 and 99).
- **The two quantities answer different questions and the difference is decision-relevant.** RAT isolates what relaxing interventions does; RAC is what a surveillance system would actually see. A criterion written on cases keeps interventions in place while the pipeline drains, even though transmission risk has already settled — and a naive model, which identifies the two by assumption, cannot represent that distinction at all.
- **Fixing k changes very little here**, as the near-identical panels show: the widest gap is 0.269 for SSE-SO and 0.236 for SSI-SO, against 0.269 and 0.240 when it is estimated. This follows from the onset-anchored k posteriors already sitting on the literature value.

11 Sensitivity analyses

None of the following has been run. They are listed in rough order of how much each would add, with what it would resolve.

1. **Outbreak-specific serial interval** (mean 19.46 d, SD 6.08 d) — *naive models only*. Its much lighter tail materially lowers the RAC after a long case-free period, and it is the estimate that should be preferred where outbreak-specific contact-tracing data exist. It is structurally inadmissible for the onset-anchored models (Section 1), and that incompatibility should be reported rather than worked around: it says something about how much structure the incubation/TOST decomposition imposes.
2. **Incubation/TOST decomposition.** Hold w fixed and re-run SSE-SO and SSI-SO under two or three other published EVD incubation estimates, letting the budget check supply the matching TOST each time. This bounds how much of the onset-anchoring result rests

on the particular split used here, which the serial interval alone does not identify. One configuration value; no new code.

3. **Filtering rather than smoothed latents.** Section 5.5 reports the signed comparison at a fixed parameter vector; repeating it integrated over the parameter posterior would turn the stated approximation into a quantified one across all three latent models.
4. **Shifted R -switch for the naive models.** The onset-anchoring comparison confounds model structure with the indexing convention for the intervention. Re-running the naive models with the switch moved back by $\mathbb{E}[A] = 11.40$ days separates them: any residual difference from the onset-anchored models is then attributable to structure rather than indexing. One configuration value.
5. **Non-empty initial incubation pipeline** for the onset-anchored models. Seed one or two already-infected individuals at day 0 with residual incubation drawn from f^{inc} , and check the effect on R_{pre} and on the RAC curve. This is the sensitivity that matters if the index case is not epidemiologically identified (Section 1).
6. **Prior sensitivity**, in particular for k , whose prior is deliberately informative and whose role differs between models.

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